B.Tech CSE - Third Year | Lab 12

Assignment 12: Recursive Factorial, Sum and Power • *Aim: Solve numeric problems using recursive methods and base cases.*

Experiment No. 12.1: Recursive factorial

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Recursive_factorial {
    static int factorial(int n) {
        if (n == 0) return 1;
        return n * factorial(n - 1);
    }
    public static void main(String[] args) {
        System.out.println(factorial(5));
    }
}
```

3. **Output:** `120`
4. **Viva-Voce:** What is recursion? What is the base case? Why is 0! equal to 1?

Experiment No. 12.2: Recursive sum from 1 to n

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Recursive_sum_from_1_to_n {
    static int sum(int n) {
        if (n == 0) return 0;
        return n + sum(n - 1);
    }
    public static void main(String[] args) {
        System.out.println(sum(10));
    }
}
```

3. **Output:** `55`
4. **Viva-Voce:** How does the recursive call get smaller? What is sum(0)? What is the sum from 1 to 10?

Experiment No. 12.3: Recursive power calculation

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Recursive_power_calculation {
    static int power(int base, int exp) {
        if (exp == 0) return 1;
        return base * power(base, exp - 1);
    }
    public static void main(String[] args) {
        System.out.println(power(2, 5));
    }
}
```

3. **Output:** `32`
4. **Viva-Voce:** What is the base case? Why is any nonzero base to exponent 0 equal to 1? How many recursive levels are used for exp = 5?